

Minimax Approximation Polynomials for Ndfloat16 Protocols

1. Inversion

```
In[*]:= ClearAll[MaxErrInv, SymmetricReciprocalPolynomial, SymmetricReciprocalNewton];
```

```
MaxErrInv[approx_, {x_, a_, b_}] :=  
  NMaxValue[{Abs[approx - 1/x], a ≤ x ≤ b}, x, WorkingPrecision → 40]
```

```
SymmetricReciprocalPolynomial[x_, degree_Integer?NonNegative] :=  
  Module[{t, residual, q},  
    residual = ChebyshevT[degree + 1, (8 t - 5) / 3] / ChebyshevT[degree + 1, -5 / 3];  
    q = Cancel[(1 - residual) / t];  
    Expand[x q /. t → x^2]  
  ]
```

```
SymmetricReciprocalNewton[x_,  
  seedDegree_Integer?NonNegative, newtonSteps_Integer?NonNegative] :=  
  Module[{seed}, seed = SymmetricReciprocalPolynomial[x, seedDegree];  
    Nest[Expand[# (2 - x #)] &, seed, newtonSteps]]
```

```
In[*]:= SymmetricReciprocalNewton[x, 2, 1]
```

Out[*]=

$$\frac{4368 x}{365} - \frac{7573056 x^3}{133225} + \frac{3653632 x^5}{26645} - \frac{23691264 x^7}{133225} + \frac{3145728 x^9}{26645} - \frac{4194304 x^{11}}{133225}$$

```
In[*]:= 
$$\frac{4368 x}{365} - \frac{7573056 x^3}{133225} + \frac{3653632 x^5}{26645} - \frac{23691264 x^7}{133225} + \frac{3145728 x^9}{26645} - \frac{4194304 x^{11}}{133225}$$

```

Out[*]=

$$\frac{4368 x}{365} - \frac{7573056 x^3}{133225} + \frac{3653632 x^5}{26645} - \frac{23691264 x^7}{133225} + \frac{3145728 x^9}{26645} - \frac{4194304 x^{11}}{133225}$$

```
In[*]:= MaxErrInv[SymmetricReciprocalNewton[x, 2, 1], {x, 1/2, 1}]
```

Out[*]=

0.01094389191217864514918371176580972039782


```
In[*]:= ClearAll[SqrtChebyshevPolynomial, SqrtChebyshevRationalPolynomial, MaxErrSqrt]
```

```
SqrtChebyshevPolynomial[
  x_,
  degree_Integer?NonNegative,
  precision_ : 50
] :=
Module[{nodes, values},

  nodes = N[
    Table[
      3 / 4 + 1 / 4 Cos[(2 k + 1) Pi / (2 degree + 2)],
      {k, 0, degree}
    ],
    precision
  ];

  values = Sqrt[nodes];

  Expand[
    InterpolatingPolynomial[
      Transpose[{nodes, values}],
      x
    ]]]

SqrtChebyshevRationalPolynomial[x_,
  degree_Integer?NonNegative, tolerance_ : 2^-14] :=
  Rationalize[SqrtChebyshevPolynomial[x, degree, 80], tolerance]
MaxErrSqrt[approx_, {x_, a_, b_}] :=
  NMaxValue[{Abs[approx - Sqrt[x]], a ≤ x ≤ b}, x, WorkingPrecision → 40]
```

```
In[*]:= data = Table[
  {i, MaxErrSqrt[SqrtChebyshevPolynomial[x, i], {x, 1/2, 1}]},
  {i, 0, 4}];
TableForm[data, TableHeadings → {None, {"i", "maxError"}}]
```

```
Out[*]//TableForm=
```

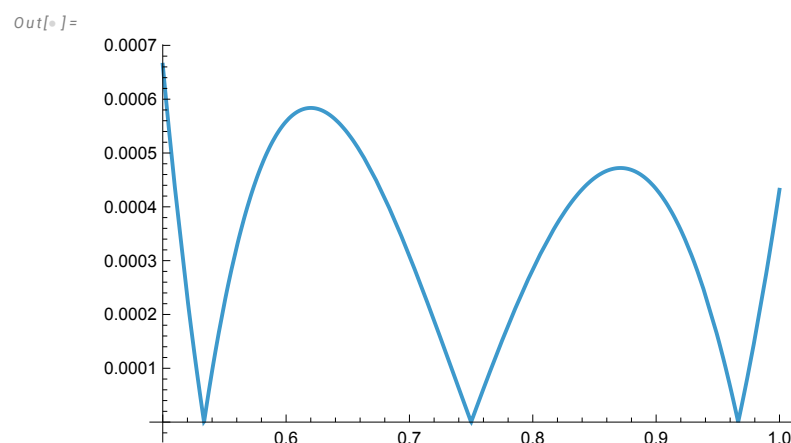
i	maxError
0	0.1589186225978911223628788086480871441866
1	0.007431885767291104998503856105482144287015
2	0.0006640275388930101891355876888007471744560
3	0.00007295416302257387294496321091576020046220
4	8.904863373085309109491032585078110039196 × 10 ⁻⁶

```
In[*]:= SqrtChebyshevRationalPolynomial[x, 2]
```

```
Out[*]=
```

$$\frac{13}{41} + \frac{191x}{217} - \frac{17x^2}{86}$$

```
In[*]:= Plot[{Abs[SqrtChebyshevPolynomial[x, 2] - Sqrt[x]]}, {x, 1/2, 1}]
```



3. Logarithm

```
In[*]:= ClearAll[Log2ChebyshevPolynomial, Log2ChebyshevRationalPolynomial, MaxErrLog2];
```

```
Log2ChebyshevPolynomial[x_, degree_Integer?NonNegative, precision_ : 50] :=
Module[{nodes, values}, nodes =
  N[Table[3/4 + 1/4 Cos[(2 k + 1) Pi / (2 degree + 2)], {k, 0, degree}], precision];
values = Log[2, nodes];
Expand[InterpolatingPolynomial[Transpose[{nodes, values}], x]]]
```

```
Log2ChebyshevRationalPolynomial[x_,
  degree_Integer?NonNegative, tolerance_ : 2^-16] :=
Rationalize[Log2ChebyshevPolynomial[x, degree, 80], tolerance]
```

```
MaxErrLog2[approx_, {x_, a_, b_}] :=
NMaxValue[{Abs[approx - Log2[x]], a ≤ x ≤ b}, x, WorkingPrecision → 40]
```

```
In[*]:= data = Table[
  {i, MaxErrLog2[Log2ChebyshevPolynomial[x, i], {x, 1/2, 1}]},
  {i, 0, 4}];
TableForm[data, TableHeadings → {None, {"i", "maxError"}}]
```

Out[*]//TableForm=

i	maxError
0	0.5849625007211561814537389439478165087598
1	0.05361838480937757651564409371515432473012
2	0.006308174563858836239494272919974729274612
3	0.0008254628229341662790276278071081185540380
4	0.0001145799603822780741549077301545441766922

```
In[*]:= Log2ChebyshevRationalPolynomial[x, 2]
```

Out[*]=

$$-\frac{719}{271} + \frac{343x}{86} - \frac{79x^2}{59}$$

